

# Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

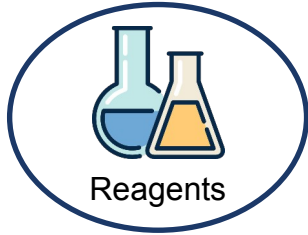


Background models

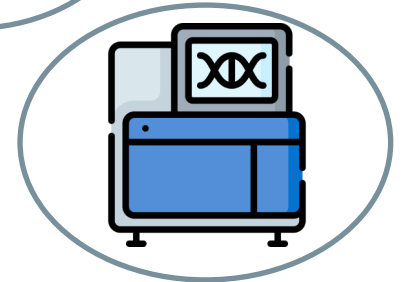
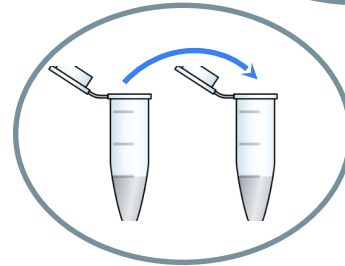
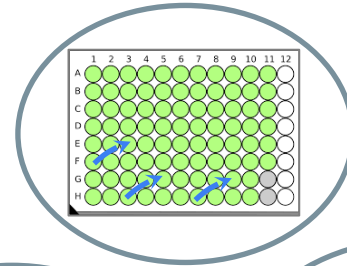
# Exploring Results

- QC metrics
- Sample Report
  - Metrics
  - Analysis
- Accounting for background

# Contamination will happen!



External  
Contamination

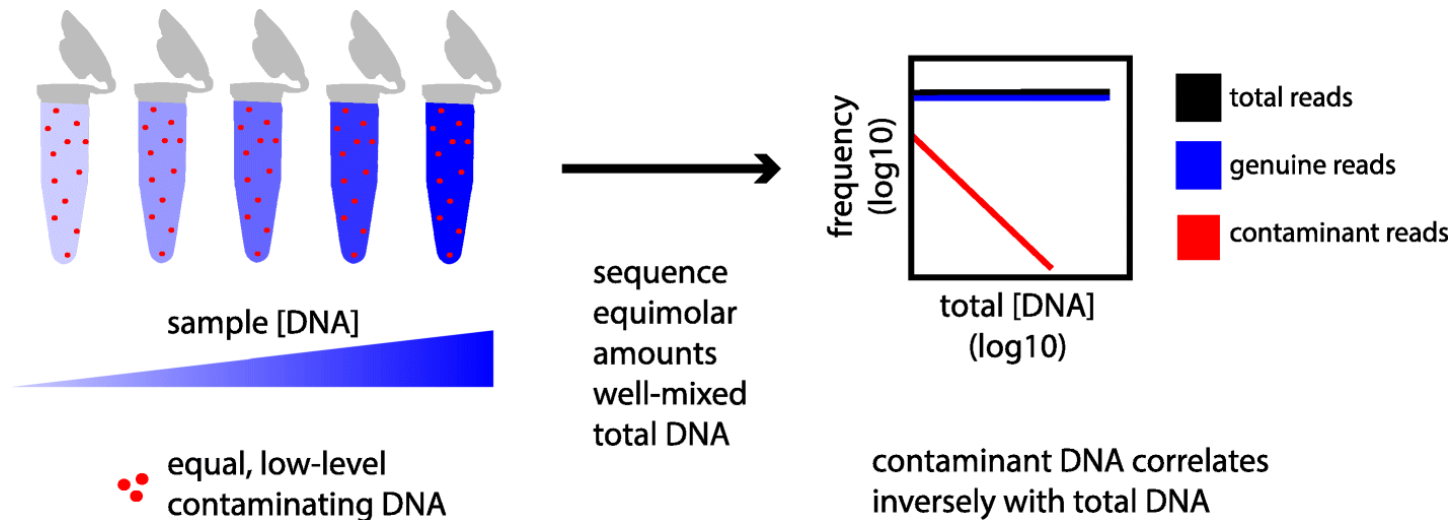


Internal  
Contamination  
(cross-contamination)

# Contamination can lead to inaccurate data interpretation

- Falsely inflates within-sample diversity
- Obscures differences between samples
- Interferes with comparisons across studies

# Contamination disproportionately affects low-biomass samples



# Water controls can help

- 'Blanks' such as water controls can help identify both internal and external contaminants
- Water controls should be processed the same way as the samples, using the the same reagents
  - Have at least 2 water controls, but the more the better!

# Background model

- Used to calculate relative abundance of the taxa found in negative controls
- Background model based on relative abundance metric:
  - Uses the bases per million (bpm) as the metric of relative abundance.

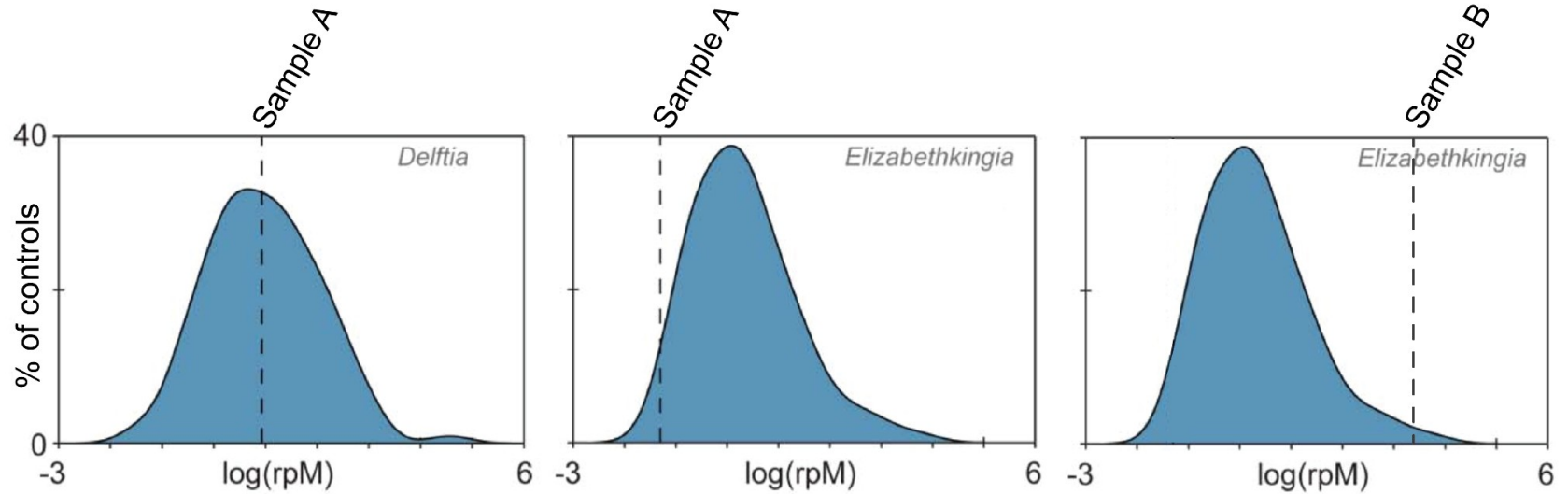


# Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
  - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?

Background distribution  
overlaid with sample taxa abundance



----- **Abundance in potentially infectious sample**

 **Distribution of abundance in controls**

# Background model enables Z-score calculation

- **Z-score**

- Measurement that describes a value's relationship to the mean of a group of values
  - How does the abundance of a taxon in a sample compare to its mean abundance in negative controls?
- It is measured in terms of standard deviations from the mean

$$Z = \frac{x - \mu}{\sigma}$$

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$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

**Z-score linear scale:**



0

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$$Z = 0$$


$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

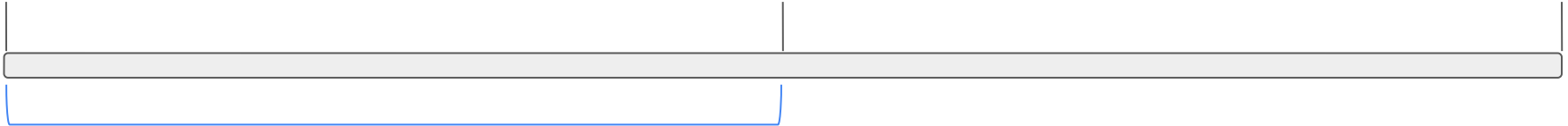
$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$Z < 0$

$Z = 0$



Taxa **more abundant in negative controls** than  
in samples

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

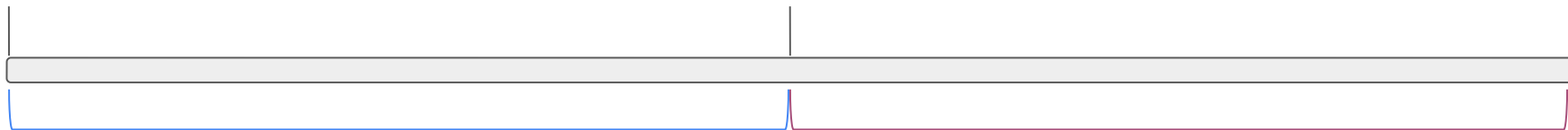
$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

No difference

$Z < 0$

$Z = 0$

$Z > 0$



Taxa **more abundant in negative controls** than  
in samples

Taxa **more abundant in samples** than negative  
controls

$$Z = \frac{x - \mu}{\sigma}$$

$x$  = taxon abundance in sample

$\mu$  = taxon mean abundance in negative controls  
(background model)

$\sigma$  = standard deviation of taxon abundance in negative  
controls (background model)

Only in  
control  
**-100**

No difference

0

Only in  
sample  
**100**

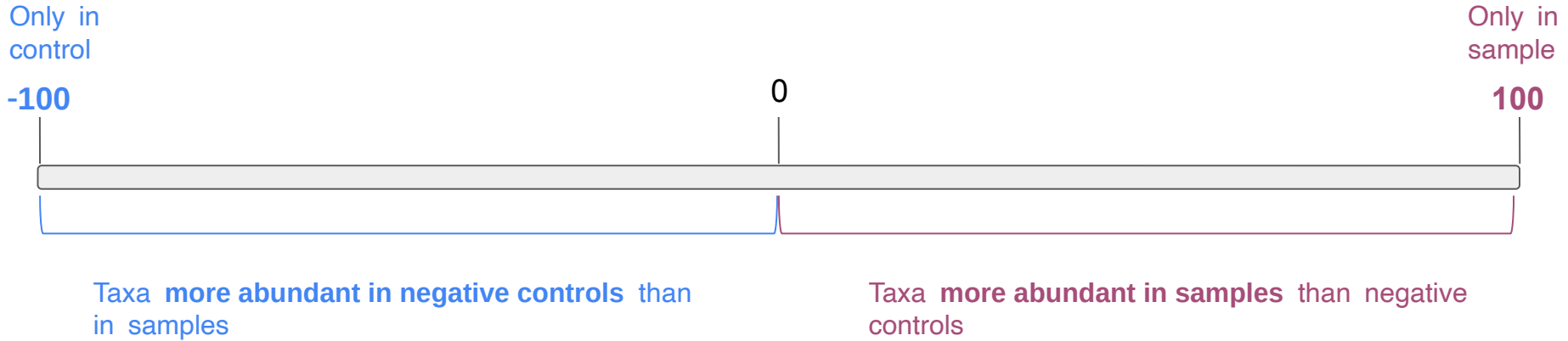
Taxa **more abundant in negative controls** than  
in samples

Taxa **more abundant in samples** than negative  
controls



# Use Z-score filter after applying background model

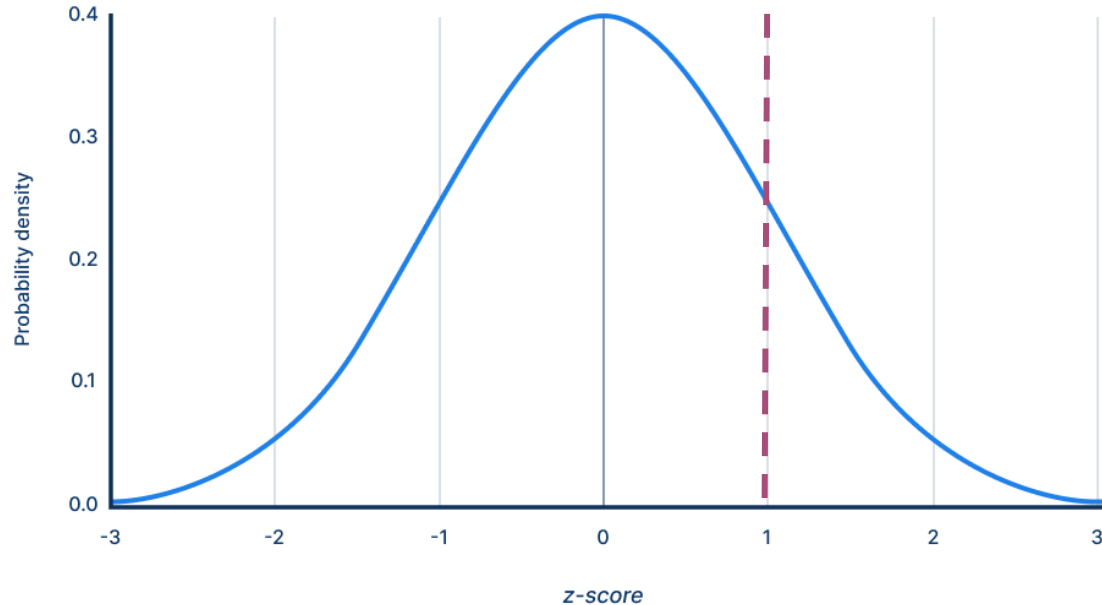
Start with taxa with Z-scores  $> 1$



# Use Z-score filter after applying background model

Start with taxa with Z-scores  $> 1$

**Standard normal distribution**



# Calculating Z-score

Standard Z-score calculation for sample “S”:

$$Z = \frac{(\text{Taxon rPM in S}) - (\text{Taxon mean rPM in Negative Controls})}{\text{Standard deviation for Taxon rPM in Negative Controls}}$$

# Additional controls

- Controls can also be used to create a background model to identify pathogens of interest (for example, cases vs controls)
- Sample controls can also be used to identify contaminants from sampling, but can be difficult to come by
- Background models can be applied to the sample report and the heatmap

# Let's make a background model!

Option 1 (outlined in worksheet): simple, subtract average signal in water controls from signal found in sample

Option 2 (shown in this slide deck): calculate a Z score for each taxa within each sample, using the equation outlined above

# Go to reports and look at the water controls

- Are your candidate taxa present in the water controls?
- If so, at what level?
- How does this compare to the level seen in your sample?

→ Subtract the background / water control signal from the sample signal for each taxa



Interpret: real signal should be higher in samples than in water controls

Example

# Example: serum sample from VHF-suspect samples

RR145011\_hac ▾

Suspected sample

Share

Download ▾

...

Nanopore

Taxon name



Name Type: Scientific ▾

Categories: 1 ▾

Threshold filters ▾

Read Specificity: Specific Only ▾

Annotation ▾

Pathogen Flags ▾

Viruses X

4 rows passing the above filters, out of 120 total rows CLEAR FILTERS



| ▾   | Taxon                                                 | bPM ▾           | b           | r       | contig | cont... | %id         | L            | E val...                | NT<br>NR |
|-----|-------------------------------------------------------|-----------------|-------------|---------|--------|---------|-------------|--------------|-------------------------|----------|
| ▾ ▢ | Orthoflavivirus ( 1 viral species: ● 1 )              | 14,096.0<br>0.0 | 56,794<br>0 | 62<br>0 | 0<br>0 | 0<br>0  | 96.5<br>0.0 | 754.7<br>0.0 | 10 <sup>-286</sup><br>0 |          |
|     | ▢ Orthoflavivirus denguei <span>Known Pathogen</span> | 14,096.0<br>0.0 | 56,794<br>0 | 62<br>0 | 0<br>0 | 0<br>0  | 96.5<br>0.0 | 754.7<br>0.0 | 10 <sup>-286</sup><br>0 |          |
| ▾ ▢ | Lentivirus ( 1 viral species )                        | 6,503.7<br>0.0  | 26,204<br>0 | 42<br>0 | 0<br>0 | 0<br>0  | 90.2<br>0.0 | 603.0<br>0.0 | 10 <sup>-247</sup><br>0 |          |
|     | ▢ Human immunodeficiency virus 1                      | 6,503.7<br>0.0  | 26,204<br>0 | 42<br>0 | 0<br>0 | 0<br>0  | 90.2<br>0.0 | 603.0<br>0.0 | 10 <sup>-247</sup><br>0 |          |



# Example: serum sample from VHF-suspect samples

Neg\_lib\_ctrl2\_hac ▾

Water control

Share

Download ▾

⋮

Nanopore

Taxon name

Q

Name Type: Scientific ▾

Categories: 1 ▾

Threshold filters ▾

Read Specificity: Specific Only ▾

Annotation ▾

Pathogen Flags ▾

Viruses X

2 rows passing the above filters, out of 77 total rows

CLEAR FILTERS

⌵

Taxon

bPM ▾

b

r

contig

cont...

%id

L

E val...

NT  
NR

▾

Orthoflavivirus ( 1 viral species: ● 1 )

19,932.9

10,850

12

0

0

96.8

758.1

10<sup>-288</sup>

▾

Orthoflavivirus denguei

Known Pathogen

19,932.9

10,850

12

0

0

96.8

758.1

10<sup>-288</sup>

# Example: serum sample from VHF-suspect samples

## Suspected sample

| bPM ∨    | b      | r  |
|----------|--------|----|
| 14,096.0 | 56,794 | 62 |
| 0.0      | 0      | 0  |
| 14,096.0 | 56,794 | 62 |
| 0.0      | 0      | 0  |

## Water control

| bPM ∨    | b      | r  |
|----------|--------|----|
| 19,932.9 | 10,850 | 12 |
| 0.0      | 0      | 0  |
| 19,932.9 | 10,850 | 12 |
| 0.0      | 0      | 0  |

# Serum library has high signal from barcode bleeding

|             |        | Sample |      | Water        |            | Sample - water |           |            |
|-------------|--------|--------|------|--------------|------------|----------------|-----------|------------|
| sample_name | Virus  | NT bPM | NT r | NT bPM water | NT r water | bPM - water    | r - water | Conclusion |
| RR145011    | Dengue | 14096  | 62   | 19932        | 12         | -5836          | 50        | Background |
|             | HIV    | 6503   | 42   | -            | -          | 6503           | 42        | Hit        |

|             |        | Sample |      | Water        |            | Sample - water |           |            |
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Subtracting background signal was necessary to help with interpretation and reduce false positives

# Example report - background subtraction necessary for accurate calls

RR145011\_hac ▾

Nanopore

Taxon name  Name Type: Scientific ▾ Categories: 1 ▾ Threshold filters ▾ Read Specificity: Specific Only

Viruses X

4 rows passing the above filters, out of 120 total rows [CLEAR FILTERS](#)

| ▼   | Taxon                                           | bPM ▾           | b           | r       | contig | cont... | %id         | L            | E val...                | NT<br>NR |
|-----|-------------------------------------------------|-----------------|-------------|---------|--------|---------|-------------|--------------|-------------------------|----------|
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This DENV is background / barcode hopping

Subtracting background signal was necessary for some samples to help with interpretation and reduce false positives

# Analysis Workflow - ONT

Complete a first pass sample report view

1. Look at the list of detected taxa
2. Record NT bPM, NT r, and any other relevant metrics
3. Look at the coverage plot for each taxa and record your interpretation (low, medium, high coverage)
4. BLASTn contigs from each candidate taxa and record what you find (specific, non-specific, etc)

Check negative water controls

1. Navigate to the sample report for each negative water control included in the run
2. For each taxa observed in the samples, record the NT bPM, NT r, and any other metrics in each water control
3. If you have >1 water control, calculate the average signal for each taxa

Put it all together (background subtraction + interpretation)

1. Subtract the background/water control signal from the sample signal for each taxa
2. Interpret: real signal should be higher in samples than in water controls

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Go to worksheet:  
Identification of  
background signal

**Worksheet:** CZ ID Training Reference Tables & Worksheets\_ONT, pg 17-20